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Equilibrium fuel supply and carbon credit pricing under market competition and environmental regulations: A California case study

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HIGHLIGHTS

- Mathematical programming with equilibrium constraints method is used.
- Investigated simultaneous equilibrium on fuel supply market and carbon trade market.
- Studied interactions between commodity market and carbon trade market.
- Cellulosic biofuels play critical role in de-carbonization transport sector.
- Mathematically and empirically proves the relationship between carbon price and regulation.

ARTICLE INFO

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Mathematical programming with equilibrium constraints (MPEC)
Carbon credit trading
Biofuel policy
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ABSTRACT

Performance-based environmental regulation has gained popularity as a policy tool to prevent climate change. California implements a Low Carbon Fuel Standard regulation to reduce the average carbon intensity of fuels by 10 percent, without specifying technologies to achieve the target. A carbon trading market is established to facilitate fuel producers making revenue by producing low carbon second-generation renewable fuels. There is a knowledge gap in understanding interactions between commodity and carbon trade markets under performance-based regulation. We propose a mathematical program with an equilibrium constraints model to find the equilibrium transportation energy portfolio under environmental protection policy. The model utilizes Karush-Kuhn-Tucker optimality conditions to represent the profit maximization of fuel suppliers. Profit is counted in both the commodity market and the carbon trading market. Our results show that carbon credit encourages the production of second-generation biofuels, which plays a critical role in the success of the Low Carbon Fuel Standard. Carbon credit price is driven by compliance with carbon intensity regulations, which we prove through mathematical formulation and empirical data analysis. Reducing carbon intensity is the key to promote biobutanol underperformance based on the low carbon fuel policy. The proposed framework, with small adjustments, can be used to evaluate performance-based regulation in other fields.

1. Introduction

The transportation sector, dominated by petroleum fuels, is a major cause of greenhouse gas (GHG) emissions worldwide [1]. In 2014, directly combusted GHG from the transportation sector contributed to 23% of energy-related GHG emissions globally [2], 26% in Europe [3], 33% in United States [4], and between 15% and 40% in major developing countries [5]. Due to concerns about air quality, energy security, and global climate change, countries around the world have developed various regulatory policies to promote the production of renewable energy to replace petroleum-based fossil fuel in transportation. For example, the European Union aimed to derive 20% of its energy from renewable sources by 2020, with 10% specifically in the transportation sector [6]. The Chinese government issued medium- and long-term plans to meet 15% of its primary energy demand with renewables by 2020 and 30% by 2030 [4]. In the United States, the national policy is the Renewable Fuel Standard (RFS) [7]. RFS requires a certain volume of renewable fuel to replace petroleum-based fuels. The target is 36 billion gallons of renewable fuels by the year 2022, of which 16 billion gallons must be second-generation biofuel, specifically cellulosic

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Table 2
Model input parameters.

	Parameter	Value	Unit	Source
Demand, M^t_G, M^t_D	Gasoline Category	13.9–14.6	Billion Gas. Gal. Eq. (GGE)	[46]
	Diesel Category	4–4.2	Billion GGE	[46]
Carbon Intensity, I_f	R_G, R_D, gasoline, diesel, electricity, H_2	20–100	gram CO_2e/MJ	[9,47]
Energy Intensity, ρ_f	R_G, R_D, gasoline, diesel, hydrogen	89–150	MJ/Gallon	[9,47]
	Electricity	3.6	MJ/kWh	[9,47]
Technology, B_f	Ethanol blend wall	15%	%	[13,16]
	Biodiesel blend wall	20%	%	[19,48]
Regulation, A^t_G, A^t_D	Gasoline Category	96–88	gram CO_2e/MJ	[9]
	Diesel Category	95–91	gram CO_2e/MJ	[9]
Production Cost, C^t_f	Gasoline	2.6–2.8	2017 $/Gallon	[49][1]
	Corn ethanol	2.2–2.3	2017 $/Gallon	[50]
	Sugarcane ethanol	1.5–1.6	2017 $/Gallon	[51][2]
	Ultra-low sulfur diesel	3–3.3	2017 $/Gallon	[49]
	Biodiesel	3.9–4.2	2017 $/Gallon	[50]
	Electricity	0.091	2017 $/kWh	[52]
	H_2	4–6	2017 $/Gallon	[53][3]
Demand Curve, $P_f = f_f(q_f)$	Each fuel type f, except hydrogen and electricity			[45,46]

[1] Production cost is derived by subtracting a historically average profit margin of the refinery industry, 14% as reported by the US Census Bureau [49], from the forecasted retail price of gasoline in the Pacific region from 2017 to 2020 [54]. This derivation process is consistent with the methodology used by CARB [9].

[2] Sugarcane ethanol is mainly imported rather than produced in the US. The production cost of sugarcane refers to import cost, for which the International Energy Agency provides trade cost information in a recent (2017) report [51].

[3] Note that NREL's estimation has a unit of $ per 1 kg of hydrogen, and the conversion factor is 1 kg hydrogen equals 1 gasoline gallon equivalent (GGE), which is the same conversion factor used in hydrogen fuel cell studies.

demand for the remaining fuels [46].

4.2. Model solution and algorithm

This model is solved by AMPL software with CONOPT solver [55]. In order to find the profit maximization of each fuel market participant and the equilibrium status, we utilized KKT transformation to convert the profit maximization problem of each fuel producer into sets of KKT optimality constraints. These constraints turn out to be quadratic rather than linear and thus cannot be solved by the basic solver in AMPL, which is only good for linear programming. CONOPT is a solver designed to solve nonlinear programming problems and can accommodate both nonlinear functions in objective and in constraint functions [56]. In the model formulation listed above, no objective is specified; rather a set of KKT optimality constraints are listed. The CONOPT solver seeks a solution point that satisfies given constraints and finds the equilibrium.

5. Results and discussion

The proposed MPEC model solves for equilibrium production for each energy supplier in the market. Each market participant's objective is to maximize its own total profit of selling energy and selling/buying carbon credit. Three ethanol producers of different resources, i.e. corn, sugarcane (importer, not producer), and cellulosic, compete with each other on the commodity market, because of the 15% ethanol blend wall in gasoline. From an economic point of view, cellulosic ethanol cannot compete with corn ethanol. But due to carbon intensity compliance and the carbon trading market, cellulosic ethanol might be competitive if profits on the two markets are counted.

5.1. Model results

Fig. 1 presents the portfolio of transportation energy and carbon intensity under market equilibrium from 2015 to 2020. One general observation is that the volume of pure gasoline (referred to as CARBOB in the regulation) gradually decreases while alternative fuels increase in order to satisfy carbon intensity compliance as defined in California's Low Carbon Fuel Standard. In the category of gasoline and gasoline substitutes, corn ethanol volume is constantly decreasing over this time period while sugarcane ethanol's volume expands, until 2018 when the trend reverses. Only cellulosic ethanol is constantly growing, with significant expansion in 2019 and 2020. When LCFS starts getting more stringent, it requires low carbon cellulosic ethanol to balance the average carbon intensity of the fuel portfolio. In the diesel and diesel substitutes category, the supply of biodiesel grows from 2015 to 2018, then plateaus after 2018. This is mainly due to the blend wall of diesel, which is 20% in the model. For carbon intensity, the results show that the model's average carbon intensity for the diesel category is constantly lower than carbon intensity compliance requirements. For the gasoline category, the modeled carbon intensity is higher than the compliance value for each year, but the gap diminishes closer to the year 2020. Clearly, the diesel market provides a large amount of carbon intensity credits to the gasoline market to meet compliance. Thus, diesel fuels are critical to the success of the Low Carbon Fuel Standard regulation.

We speculate some explanations for this market equilibrium fuel portfolio. The transportation energy demand is in the unit of gasoline gallon equivalent (the same unit as defined in California's EMFAC and Vision 2.1 model), but LCFS's average carbon intensity is based on the unit of gram CO_2e per MJ. Therefore, a fuel supplier's profit maximization and its equilibrium will take into consideration the cost per gasoline gallon equivalent of each type of fuel as well as the energy stored in each gallon of fuel, i.e. energy intensity. Since CARBOB has the lowest production cost per MJ, the model chooses to use as much as possible as if there were no carbon compliance. However, as compliance value gets lower over time, low carbon biofuels (sugarcane or cellulosic ethanol) gradually play important roles in the fuel pathway. Corn ethanol has the lowest cost but a carbon intensity value as high as that of CARBOB. Cellulosic ethanol has the lowest carbon intensity but highest cost because it is still a new technology. Sugarcane ethanol's cost and carbon intensity are in the middle compared to the other two biofuels. Therefore, it is expected that corn ethanol is always preferred in the fuel portfolio whenever environmental constraint is not a concern. When carbon compliance starts to be implemented but not quite stringently, sugarcane becomes attractive because of its relative low cost and moderate carbon benefit. Finally, when compliance is stringent, such as after 2018, a large volume of cellulosic ethanol has to be produced to balance carbon intensity to meet compliance. This explanation is consistent with observations in the model results as shown

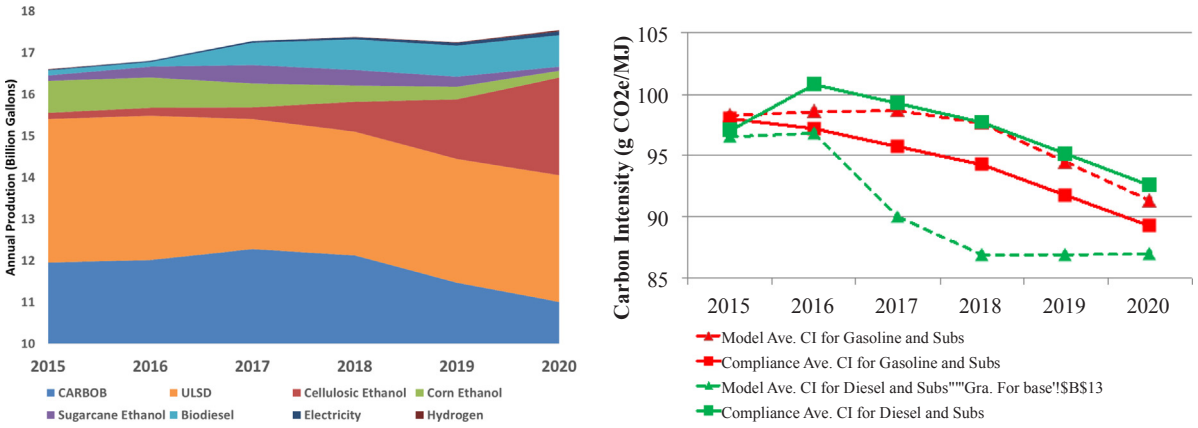

Fig. 1. Transportation energy portfolio (left) and carbon intensity (right) under equilibrium.

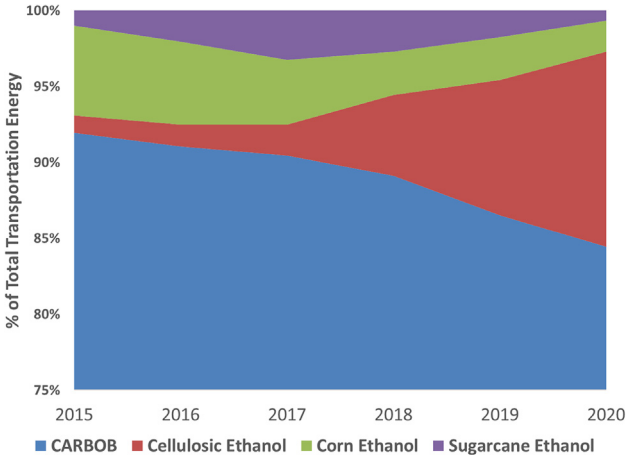

Fig. 2. Portfolio of gasoline and gasoline substitutes.

in Figs. 1 and 2.

5.2. Model validation with actual market data

We compared our modeled fuel portfolio with actual fuels supplied in the California market from 2015 to 2017 in Fig. 3. The actual market data is from Yeh et al. [12]. We observed that our model's results correctly predict the trend of biofuels but the magnitude of our prediction is smaller than the actual values that emerged in the market. For example, in our simulation, corn ethanol's volume decreased from 760 to 670 million gallons from 2015 to 2017. In the actual market, corn ethanol experienced a similar decreasing trend, but the exact number is larger than the numbers in our paper [12]. Similar findings are observed for sugarcane and biodiesel. One possible explanation is that, in our model, carbon credits can be used only in the year they are generated. However, in reality, fuel suppliers are allowed to bank carbon credits for future use or trading. In the early years of LCFS regulation (2011–2013), a large amount of carbon credit was generated because of the high carbon intensity compliance [12]. This allows the market to produce more low cost but high carbon intensity CARBOB and less biofuels in later years. But increasingly stringent carbon intensity compliance drives the supply of low carbon biofuels to grow. This matches the observed difference in magnitude and similarity in trend between model estimation and actual market performance.

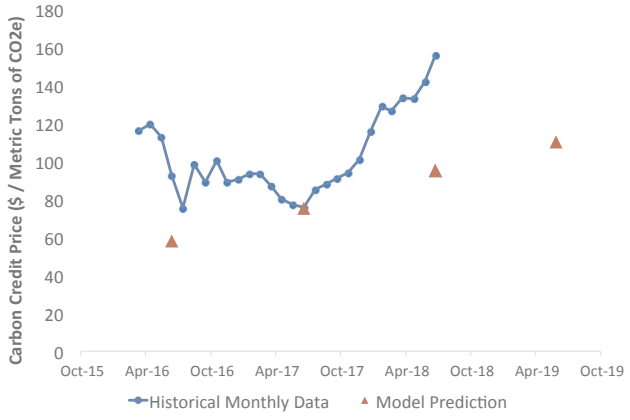

Fig. 3. Historical vs. model-predicted carbon credit trade price.

We compare our modeled carbon credit price with actual market price in Fig. 3. In the real carbon trade market, the price fluctuated around \$90 per metric ton of CO_2e from 2016 to the end of 2017, then rapidly climbed up to around \$130 for the first half of 2018, and finally reached around \$155 per metric ton of CO_2e in June 2018. Our model predicts a similar trend but with lower magnitude. Specifically, our model predicts the carbon price to be around \$60 per metric ton of CO_2e for 2016 and 2017, then \$100 in 2018 and \$120 in 2019. The consistency in trends between historical data and model prediction is expected, because a major driver of carbon credit price is carbon intensity compliance. The average carbon intensity compliance is at 98 g CO_2e/MJ from 2013 to 2015, and then the compliance rate is reduced by 1.5% each year from 2016 to 2018 and 2.5% for 2019 and 2020 respectively. The variation between actual and predicted carbon price is consistent with the increasingly aggressive compliance regulation. It is worth noting that carbon price was fluctuating around \$90 per metric ton of CO_2e for 2016 and most of 2017, even though 2016 is the year carbon intensity compliance started to decrease. Our explanation is that a large number of carbon credits were generated before 2015, when compliance was easy to meet (this was confirmed by Yeh et al. [12]). These carbon credits were banked and began being traded during 2016 and 2017, which kept the carbon price around \$90. However, since our model does not allow the banking of carbon credit, it predicts that the carbon credit price will continuously increase when compliance becomes more stringent at a similar rate. There might be various reasons for our model predicted carbon credit price lower than actual values. This will be topics for future studies to investigate detailed explanations and further improve our model.

5.3. Comparison with system optimal approach

Equilibrium approach is a useful way to investigate market competition and performance under environmental regulations. Another is system approach. This approach assumes a system planner who can control the production of each energy supplier. The purpose of the

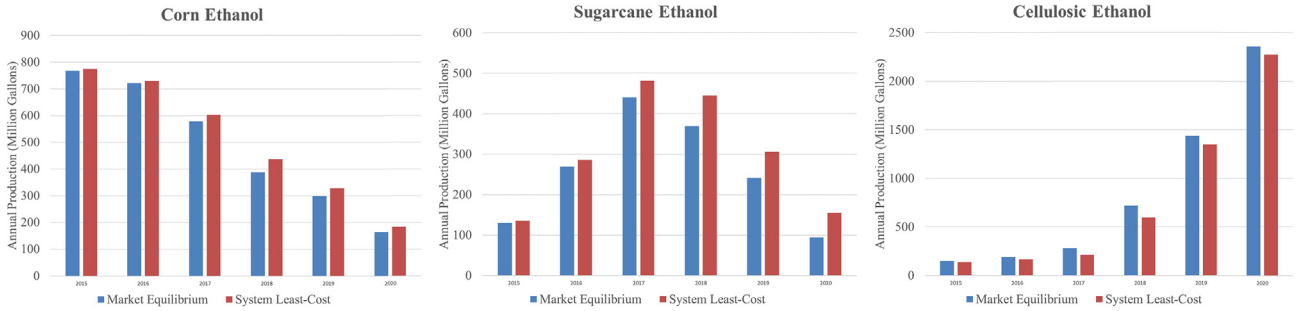

Fig. 4. Comparison of corn (left), sugarcane (middle), and cellulosic (right) ethanol production in market and system cases.

system planner is to provide transportation energy while satisfying environmental responsibility using a least-cost fuel portfolio. This least-cost solution considers only the production cost of transportation energy because production cost is the only cost to the system. Whether commodity trades between consumers and suppliers or carbon credit trades between suppliers, these are zero-sum trades. Noting that, we do not consider consumer surplus or production surplus in this study, but rather focus exclusively on explicit monetary costs.

We compare transportation energy portfolios under the system least-cost case and the market equilibrium case. Fig. 4 presents the production levels of corn (left), sugarcane (middle), and cellulosic (right) ethanol in the two cases. The system least-cost approach leads to less production of cellulosic ethanol. The difference tends to enlarge initially and then diminish after the year 2019. Here are some of our explanations. Though cellulosic ethanol has the lowest carbon intensity among all fuels, it also has the highest production cost in terms of volume ($ per gallon) or energy ($ per MJ). Thus, the system planner avoids producing cellulosic ethanol unless to satisfy environmental compliance. But in the market equilibrium case, the producer of cellulosic ethanol can obtain revenue from both the commodity and carbon trading markets. It can produce even when its marginal production revenue is below marginal cost, because the negative revenue is offset by profits from carbon credit trading. This explains the observation that cellulosic ethanol is produced at a higher level in the market equilibrium case than in the system case. As compliance gets more aggressive, such as in the years 2019 and 2020, environmental constraint as well as the blend wall constraint (15% ethanol blend rate) become the major constraints for both cases. Hence, as observed in the two cases, the majority of that 15% ethanol must come from cellulosic in order to meet compliance.

It is also observed that more sugarcane ethanol is imported in the system case between 2015 and 2017. Sugarcane ethanol has higher carbon intensity than cellulosic, but is still 30% lower than compliance. In order to achieve least production cost, the system planner used sugarcane ethanol as an interim solution during 2015–2017, before cellulosic ethanol dominated the ethanol market after 2018. But in the market equilibrium case, sugarcane ethanol faces competition from cellulosic ethanol because of the higher carbon credit profit cellulosic ethanol can generate with the same volume of production. The production quantity gap between the two cases gradually decreases after 2018. A similar pattern is observed for corn ethanol. In both cases, corn ethanol production is gradually reduced because of its high carbon intensity. In the system case, corn ethanol is produced at the maximum as long as the portfolio's average carbon intensity is maintained at compliance level. But in the market equilibrium case, corn ethanol's high carbon intensity means less carbon credit revenue; thus corn ethanol is produced at lower levels to maintain a higher selling price of corn ethanol for profit maximization.

Through the above comparison, we find that the carbon credit market encourages more production of low carbon intensity fuel, i.e. cellulosic ethanol, by internalizing the environmental externality of transportation fuels.

5.4. Implications of biobutanol on the transportation fuel portfolio

In the literature review section, we acknowledged that this study mainly focuses on bioethanol and biodiesel, which are expected to play major roles in low carbon transportation fuel policies in the U.S. [12], Canada [38], the European Union [39], and major developing counties [40]. Biobutanol is a promising biofuel with advantages and disadvantages. In this subsection, we assess implications of biobutanol on the transportation fuel portfolio. Specifically, in our model, we add biobutanol as one of the gasoline substitute fuel types, which makes it possible to be blended with CARBOB (pure gasoline) with the 16% volume ratio limit recommended by U.S. Environmental Protection Agency. The energy density of biobutanol is 110.53 MJ/gallon [27], which is lower than the energy density of gasoline, but higher than those of bioethanol. The carbon intensity of biobutanol needs to be based on life cycle analysis. Wu et al. [57] evaluated the life cycle energy and GHG emissions of biobutanol as a transportation fuel. Based on various production processes, its GHG emissions are different. Without losing generality, we choose the reported average value, i.e. 108.83 g CO_2e/MJ, as the carbon intensity for biobutanol. This is significantly higher than the carbon intensity values of bioethanol (20.4, 69.5 and 91.0 g CO_2e/MJ for cellulosic, sugarcane and corn ethanol), and higher than that of gasoline, 95.86 g CO_2e/MJ. Butanol is still not at the stage of industrial scale mass production; thus its production cost is high and its future is uncertain [58]. Currently, butanol is selling at $1–2 per kg,[3] which converts to $3.1–6.2 per gallon, based on its volumetric density. Assuming a 25% profit margin for the refinery industry (which is higher than the 14% profit margin for the petroleum refinery industry), we build two butanol cost scenarios at $2.3 and $4.7 per GGE. For the demand curve, since very little demand data exists for butanol, we assume that its demand curve has a form similar to that of cellulosic ethanol (since both are new technologies) but adjust the demand curve parameters based on the cost difference of the two biofuels. We acknowledge that there might be more suitable values for the cost and demand parameters of butanol, but given its uncertainties on the biofuel market, and since the objective of this study is focusing on building a novel mathematical model, we can improve the parameters when more information and knowledge are available.

We show the biofuel portfolio results in Fig. 5. Because current developments mainly use biobutanol as a gasoline additive with the recommended 16% limit, we present only the portfolio of gasoline substitute fuels, i.e. bioethanol and biobutanol. The results show that, in the $4.7 per gallon cost scenario, there is no biobutanol production. One advantage of butanol is its high energy density, which is 45% higher than that of ethanol. The model has an energy content demand (in MJ) constraint; thus all other things being equal, it will prefer fuel with higher energy density. But when butanol's price is at $4.7 and its carbon emission is also higher than those of bioethanols, the model decides not to produce any butanol, and the fuel portfolio will be the

[3] Based on a search on the butanol (mainly N-butanol) trade market.

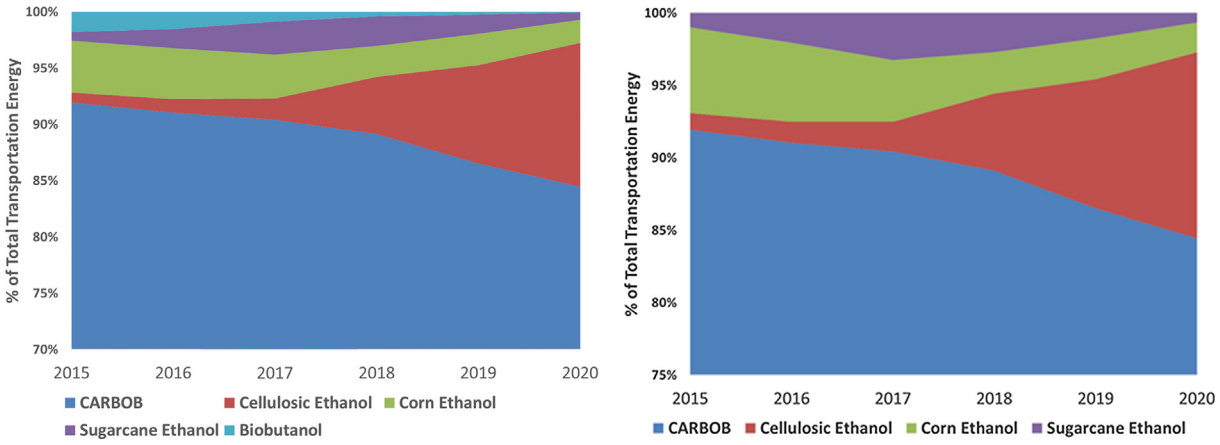

Fig. 5. Portfolio of gasoline and gasoline substitutes with biobutanol production costs at $3.1 (left) and $4.7 (right) per gallon.

same as shown in Fig. 2 (the case when butanol is not a fuel option). When butanol cost is $3.1 per gallon, it obtains some market share in the early years (2015–2017) and diminishes after 2018 when low carbon regulation is tightened. This result is expected. Butanol has high energy density and lower cost per MJ content; thus it gains advantages in the early years when the carbon intensity requirement is less stringent. Butanol's high carbon intensity leads to a carbon deficit and associated costs; therefore, after the year 2018, it diminishes.

5.5. Policy implications

We summarize some policy implications based on the modelling and findings of our study. It is worth noting that the implications are not only applicable to the Low Carbon Fuel Standard in California, but are also helpful to policy makers in other regions who are planning or evaluating the effectiveness of performance-based renewable energy policies to achieve environmental targets.

First, our results show that diesel fuels, particularly bio-based diesel fuels, are critical for successfully achieving a sustainable transportation energy portfolio. As shown in Fig. 1, the diesel fuel category provides the majority of carbon credits to satisfy average carbon intensity compliance. Specifically, the percentage of biodiesel blended into diesel fuel is projected to increase from 4% in 2015 (this was our model's prediction and actually observed in California [12]) to 20% in 2020, noting that 20% is the maximum share it can reach due to the biodiesel blend wall. It shows that the average carbon intensity for the diesel category is always lower than its carbon intensity compliance, but for the gasoline category this ratio is reversed. This difference calls for additional policies or regulations to facilitate advanced biodiesel production and usage. Particularly, most diesel engine warranties limit the use of diesel fuel containing less than 5% biodiesel [48]. Without lifting this warranty limit, biodiesel's large-scale adoption cannot be realized, and so neither can the environmental targets in transportation energy. Therefore, government efforts are needed to facilitate the certification of diesel engine performance running on diesel fuel with a high percentage of biodiesel.

Second, an appropriately-designed financial instrument can promote the production of low carbon technologies with robust market participation. In this study, we prove, through mathematical inference and empirical comparison, that there is a strong linkage between financial instrument, i.e. carbon credit price, and regulation, i.e. carbon intensity compliance, within a performance-based regulation framework. Through its designed financial instrument and market mechanism, this performance-based framework promotes the production of second-generation biofuel, cellulosic ethanol, which was not achieved by the technology-based regulation of the Renewable Fuel Standard. A similar relationship between financial instrument and regulation is also observed in other fields, such as in the power utility industry [59]. Therefore, modelling work is needed during the policy preparation stage to better design policies for achieving sustainability targets using a market approach.

Thirdly, biobutanol, as a promising biofuel, has high energy density (low cost per MJ energy content) as well as high carbon emissions per MJ. These features make it competitive when carbon intensity regulation is less strict. The low carbon regulation and carbon deficit cost will force butanol out of future fuel portfolios. Therefore, reducing carbon intensity is the key to promote butanol under a performance-based low carbon fuel policy. Some recent literature [60–62] analysed the life cycle of GHG emissions of butanol using various innovative processes and raw materials. If successful, biobutanol could be a promising biofuel in the future of the transportation sector.

6. Conclusions

The main contribution of this study is to establish a mathematical program with an equilibrium constraints model to investigate transportation energy market equilibrium under performance-based environmental regulations. Performance-based environmental regulation refers to policies such as California's Low Carbon Fuel Standard, which requires energy providers to reduce incrementally the carbon intensity of their fuels with any format of energy portfolio, rather than setting specific technological targets such as the Renewable Fuel Standard. In this study, the market equilibrium solutions are found by transferring each energy provider's profit maximization to Karush-Kuhn-Tucker optimality conditions. Profits through commodity trading and carbon trading are considered. This is the first study to simultaneously model the commodity market and the carbon trading market under performance-based environmental regulation. The findings of this study can help researchers and policy makers design performance-based regulations and evaluate the effectiveness of the carbon credit trading mechanism.

Our study confirms that reducing transportation greenhouse gas emissions requires a portfolio approach. The carbon credit trading market plays a critical role to economically promote the production of high cost, low emission, second-generation biofuel (i.e. cellulosic ethanol). The diesel fuel category, particularly biodiesel, is a major driver for reducing carbon emissions. In the gasoline fuel category, the annual production volume of cellulosic ethanol has to reach around 2 billion gallons by the end of 2020 in order to satisfy the most stringent compliance under the Low Carbon Fuel Standard. Although its production cost is the lowest among biomass-based fuels, corn ethanol's production volume is gradually reduced because of its high carbon intensity value. Sugarcane ethanol imports first increase then decrease after 2017, which proves its role as the interim solution to satisfy low

carbon fuel policy. The modeled carbon credit price is estimated to increase from around $60 per metric ton CO_2e in 2015 to $120 in 2019. The trend of carbon credit price is driven by carbon intensity compliance.

We validate our model results with actual data. Our model results, both for fuel production quantity and carbon credit price, match the trend of variation in actual data. However, the values fail to match exactly. One explanation is that our model does not allow carbon credit banking for use in future years, which is allowed in the actual market. This can impact the supply of carbon credit as well as its price. Other factors, such as carbon trade transaction costs, could also be responsible for the inconsistency of our model results with actual data.

We compare transportation energy portfolios under system least-cost and market equilibrium cases. The market equilibrium case produces more cellulosic ethanol than the system case, particularly for the early years. Implementing the carbon credit market encourages the production of cellulosic ethanol because of its potential revenue on carbon credit trading. In the meantime, less corn ethanol and sugarcane ethanol are produced/imported in the market equilibrium case compared to the system case. This is expected given that corn, sugarcane and cellulosic ethanol are competing for the 15% volume mandated to gasoline suppliers by the biofuel blend wall rule.

Our findings also lead to some policy implications. First, diesel fuels have great potential in de-carbonization and are critical to achieving a sustainable transportation energy portfolio. The majority of heavy-duty freight vehicles are using diesel fuels. It is easier to utilize government efforts to promote advanced biomass-based diesel fuels across the business sector compared to the passenger vehicle fleet, which involves distributed individual choices. Second, there is a strong link between financial instrument, i.e. carbon credit price, and regulation, i.e. carbon intensity compliance, within performance-based regulation. Detailed modelling work is needed during the policy preparation stage to better design policies for achieving sustainability targets using the market approach. Third, biobutanol is a promising biofuel in future transportation energy portfolios. Reducing carbon intensity is the key to promote biobutanol under performance-based low carbon fuel policies.

Future research can focus on relaxing some assumptions or constraints developed in this market equilibrium modeling framework. Examples are:

(1) Allowing the banking of carbon credits for future trading or using other financial instruments to fully leverage the advantages of advanced low carbon biofuels in order to encourage its production;
(2) Having multiple fuel suppliers and adding spatial characteristics of fuels into our model. The production location of biofuel significantly influences its carbon footprint. Adding these spatial features can improve the representativeness of our model to the real market.
(3) Investigating government control mechanisms, such as provide carbon price guidance, to maximize social welfare. Clearly, the current market equilibrium model is not system optimal. Methods should be developed to help the government manage and guide renewable energy policies to achieve optimal social welfare.

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